The opioid system and food intake: homeostatic and hedonic mechanisms.

. . . . .

Opioids are important in reward processes leading to addictive behavior such as self-administration of opioids and other drugs of abuse including nicotine and alcohol. Opioids are also involved in a broadly distributed neural network that regulates eating behavior, affecting both homeostatic and hedonic mechanisms. In this sense, opioids are particularly implicated in the modulation of highly palatable foods, and opioid antagonists attenuate both addictive drug taking and appetite for palatable food. Thus, craving for palatable food could be considered as a form of opioid-related addiction. There are three main families of opioid receptors (μ, ■, and δ) of which μ-receptors are most strongly implicated in reward. Administration of selective μ-agonists into the NAcc of rodents induces feeding even in satiated animals, while administration of μ-antagonists reduces food intake. Pharmacological studies also suggest a role for ■- and δ-opioid receptors. Preliminary data from transgenic knockout models suggest that mice lacking some of these receptors are resistant to high-fat diet-induced obesity.